



Fig. 5: Vehicle-to-vehicle communication rendering (Source: wot.motortrend.com)



Fig. 6: BMW i8's new headlights, from LED low beams (left), shining 100m ahead (middle) to LED high beams (right) (Source: <http://ecomento.com>)

so that it is available whenever a diagnosis is done later. The car repairs shop must have access to information either in hard copy form or online to translate what the codes mean and how to go about diagnosing the particular problem.

Technological advancements have made more auto repairs possible. Generally, car computers remind you to check the tyres pressure, coolant level and see if the brake oil and engine levels are adequate. Some computers even remind you when a service or insurance premium is due.

In time to come, your car will even be able to tell you if the BMW in front is facing some problem. This will help you decide whether you should stay clear of it to prevent an accident or you should extend some help to the driver.

Soon you should be able to up-

load your car computer diagnostics data to a website and analyse it for some recommendations on how to get your car back on the road. And if you are not the do-it-yourself type, you will be able to send the data

to a repairs shop wirelessly. The shop could then send you an estimate with a list of all the required jobs that need to be done on your car. You could simply accept or reject the offer, without having to go physically all the way to them.

Perhaps our great grand children will not have to worry about maintenance of their cars at all. Just as cars would drive themselves, these would also be able to link up with the on-board diagnostics to see if there are issues and, if so, self-drive to the repairs shop.

The road ahead

As the degree of computerisation in cars increases these will be able to exchange info about traffic, weather, road conditions and several other aspects. However, one of the most important aspects is that, cars will be able to com-

municate their velocity and direction and caution each other about accidents that could happen. This technology is called vehicle-to-vehicle communication, or V2V, and we can look forward to it soon.

On-car devices equipped with V2V emit a short-range safety signal ten times per second and detect signals from other vehicles to determine whether a potential accident is about to happen. Cars equipped with the devices will sound beeps when they detect possible hazards, such as another vehicle entering an intersection, an approaching pedestrian, a patch of slippery ice ahead, or even their driver speeding too fast around a curve.

Then we have researchers, app developers and car companies developing technologies to scrutinise human drivers in a manner that ensures there are no accidents. Advanced sensors in the passenger cabin will keep an eye on a driver's key parameters, such as heart rate, eye movements and brain activity, to sense any abnormal condition—be it sleepiness or a heart attack.

In-car systems are increasingly being geared up to operate in synergy with smartphones and tablets. Chevrolet, Honda and other car brands have joined Apple to equip cars with an eyes-free mode for Siri—the voice assistant on an iPhone. The system enables drivers tell Siri to send messages, create calendar events or activate turn-by-turn navigation, without the need to take their mobile phone in their hands.

Next, we have car manufacturers innovating a new generation of smart headlights that can automatically regulate their brightness or direction based on on-road conditions. Laser high beams will be used to light up roads for nearly half a kilometre—which is twice the range of LED high-beam headlights and using even lesser energy.

We are not quite sure what the future holds for the auto industry in the long run. We might see robots sitting behind the steering wheels and driving our fully automated cars! ●

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